



The Great Seed Germination Race!

My Fun Science Discovery Workbook

Grade 2 Science Lab

Hello, Little Scientist! 🧐

Have you ever seen a tiny seed wake up and start to grow? It is just like magic! In this workbook, we will explore how seeds live, look at the three magic things they need to grow, and see who wins the **Great Sprouting Race!**

Name: _____ Date: _____

What is Germination?

Seeds look like they are deep asleep, but they are actually alive! **Germination** is the special scientific word for when a sleeping seed wakes up and begins to grow into a brand-new baby plant.

Seed Coat

The protective outer shell covering a seed.

Like a cozy warm blanket wrapped around a baby!

Cotyledon

The first inner leaves that hold food for the seed.

Like a packed lunch box inside the seed!

Sprout

A tiny, freshly awoken newborn plant.

An **Ankurit Moong** is a delicious example from breakfast!

The 3 Magic Conditions Needed

To wake up a sleeping seed, you don't need magic spells—you just need three simple things:

1. Water

Seeds drink water to puff up like a sponge and soften their hard coat.

2. Warmth

Seeds need cozy, warm surroundings to stay active and happy.

3. Air

Seeds breathe air through small hidden spaces inside the loose soil.

🎵 **Sing and Remember!** 🎵

***"Water, warmth, and air,
Three things seeds need to grow out there!"***

The 5 Adventure Stages of a Seed

Watch how a seed goes from a cozy sleeper into a proud little seedling:

STAGE 1

Drinking and Swelling

The seed drinks water and swells up big and soft, exactly like a sponge!

STAGE 2

The Blanket Breaks

Pop! The protective outer seed coat cracks open, and a little white tip peeks out.

STAGE 3

Root Digs Down (Radicle)

The very first root grows **downward** into the ground. These are the plant's drinking-straw feet!

STAGE 4

Shoot Reaches Up (Plumule)

A tiny green stem pushes **upward** toward the sun, like stretching your arms high in the morning!

STAGE 5

Hello, First Leaves!

The baby leaves open wide to catch sunlight, helping the new seedling make its own fresh food.

The Action Rhyme

*"Water wakes it, coat breaks it,
Root digs down, shoot rises up,
Leaves open wide—a seedling grows!"*



The Great Germination Race!

Not all seeds sprout at the same speed! Let's see who wins the race track:

Seed Character	Sprouting Time	Seed Coat Thickness	Race Finish Speed
 Moong Seed	1 - 2 Days	Thin Outer Coat	 1st Place (Fastest!)
 Chana Seed	2 - 3 Days	Medium Coat	 2nd Place (Medium)
 Wheat Seed	2 - 4 Days	Thick Outer Coat	 3rd Place (Slowest)



Fast Fact!

Because **Moong** has a super thin coat, it drinks water the fastest! That is why grandmothers soak moong seeds overnight to make healthy, crunchy breakfast sprouts (*Ankurit Moong*).



My Home Sprouting Journal

Try it at home! Soak some moong seeds in water overnight, drain them, and wrap them in a damp cloth. Write or draw what you see each day:

Day	What I See (Words)	My Sketch
Day 1	The seeds are in water...	
Day 2		
Day 3		

Day	What I See (Words)	My Sketch
Day 4		



Fun Science Practice Worksheet

Part A: Fill in the Blanks!

1. Germination means when a sleeping seed _____ up and starts to grow.
2. The three magic things seeds need are water, _____, and air.
3. A seed coat is like a warm, cozy _____ wrapped around the baby plant.
4. The seed that wins the germination speed race is the _____ seed!

Part B: True or False?

- ☐ **True** ☐ **False** — Seeds need sunlight while they are underground in the dark dirt.
- ☐ **True** ☐ **False** — The first root (radicle) always grows downward into the ground.
- ☐ **True** ☐ **False** — All seeds sprout at the exact same speed.

Part C: Put the Stages in Order (Write numbers 1 to 5)

- [] Tiny green leaves open wide to catch sunlight.

- [] The seed drinks water like a sponge and swells up plump.

- [] A little green stem shoots upward toward the sky.

- [] The protective outer coat cracks and splits open.

- [] A white baby root digs downward into the dark soil.

 **Congratulations! You are now an Official Seed Scientist!** 